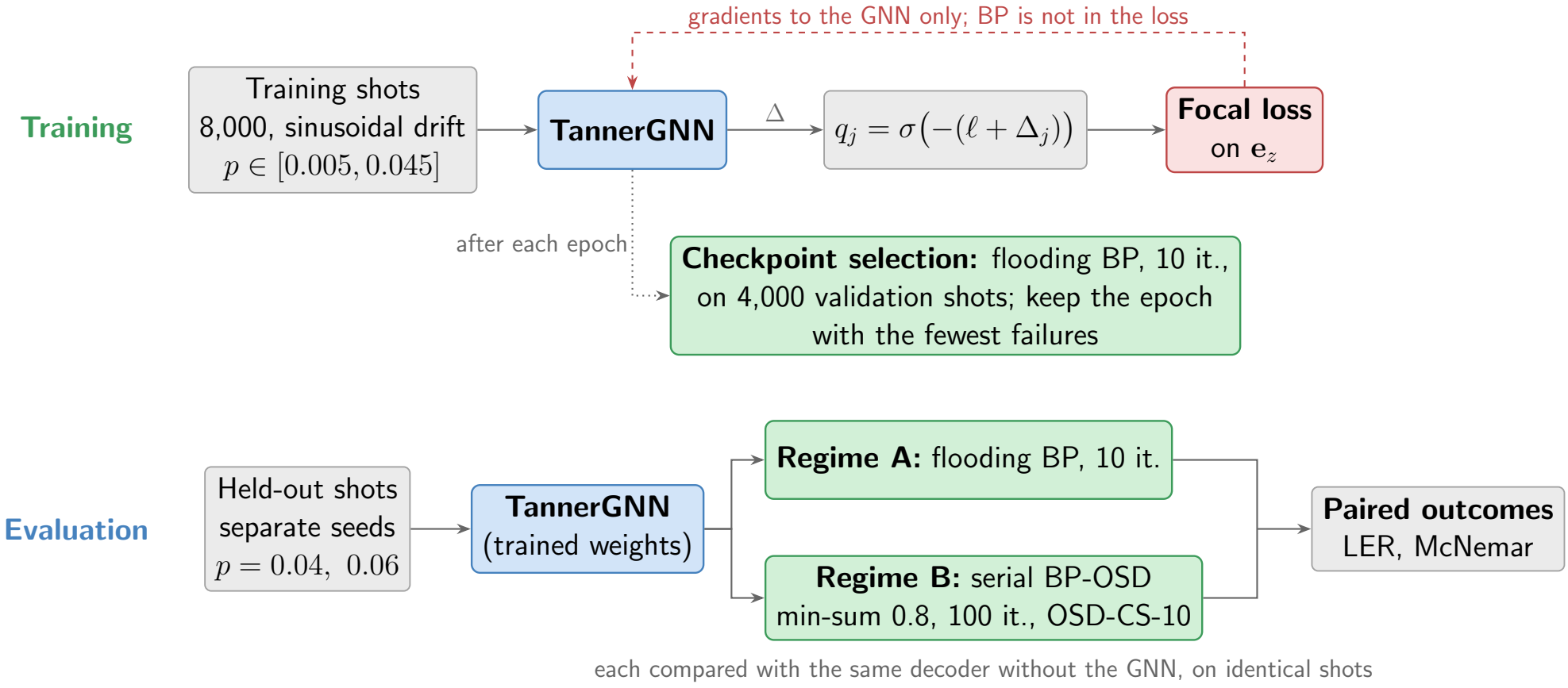

gradients to the GNN only; BP is not in the loss
Training
Training shots
8,000, sinusoidal drift
$p \in [0.005, 0.045]$
TannerGNN
Δ
$q_j = \sigma\big(-(\ell + \Delta_j)\big)$
Focal loss
on $\mathbf{e}_z$
after each epoch
Checkpoint selection: flooding BP, 10 it., on 4,000 validation shots; keep the epoch with the fewest failures
Evaluation
Held-out shots
separate seeds
$p = 0.04,\ 0.06$
TannerGNN
(trained weights)
Regime A: flooding BP, 10 it.
Regime B: serial BP-OSD
min-sum 0.8, 100 it., OSD-CS-10
Paired outcomes
LER, McNemar
each compared with the same decoder without the GNN, on identical shots